MicroPulse — Decel Intelligence Guide

Reading the McBurnie 2022 deceleration framework

MicroPulse Performance Intelligence Platform

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1. What this page tells you

The Decel Intelligence page (Monitoring → Decel Intelligence) is the most predictive single view MicroPulse offers for soft-tissue injury risk. It is built on the **McBurnie 2022** framework — the most widely cited modern model for how high-intensity decelerations contribute to muscle damage in football.

Decelerations are the single most damaging movement pattern in field sport. A high-speed sprint loads the muscles eccentrically far more than the equivalent acceleration. McBurnie's contribution was showing that the *ratio* between decelerations and other movement patterns matters more than the raw deceleration count in isolation, and that five separate ratios — taken together — explain the majority of the variance in soft-tissue injury risk.

This page converts your Catapult IMA and GPS bands into those five ratios for every player on your roster, every day, and tells you who needs attention before they get hurt.

> **Sport-science background**
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> McBurnie, Harper, Jones, Bramah and others have published extensively on this framework since 2019. The version in MicroPulse implements the full 2022 update with the indoor/outdoor sprint denominator switch added in 2024. References are listed in the methodology footer at the bottom of the page.

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2. The five dimensions

Every player gets a score on five separate dimensions. Each dimension is reported as **GREEN** (within personal range), **YELLOW** (drifting), or **RED** (clear flag).

2.1 Overload — "Has decel volume been climbing?"

Compares the player's last 28 days of decel count against their personal 28-day rolling baseline. A player whose decel volume has climbed substantially above their own normal range gets flagged here, even if their absolute number is moderate.

Acts on this when: cumulative decel exposure is creeping up over weeks. Catches gradual progressive overload.

2.2 Underload — "Are they ready for match demands?"

Compares the player's last 7 days of decel count against the team's typical match-day demand. A player whose training-week deceleration load is far below match demand will struggle on matchday — this is the *under*-stimulation flag.

Acts on this when: a player has been protected too long (post-injury, post-illness, light training week) and risks being thrown into a match without being conditioned for the demand.

2.3 A:D Coupling — "Are accelerations and decelerations balanced?"

Compares high-intensity (Band 3) accelerations against high-intensity (Band 3) decelerations. A healthy player produces roughly 1:1 — every hard sprint is matched by a hard stop. When this drops below ~0.7 it usually means the player is *braking more than accelerating* — a fatigue marker that often precedes hamstring issues.

Acts on this when: ratio is under 0.7. The player is doing more eccentric work than the system is built for.

2.4 D:Sprint Coupling — "How many decels per sprint?"

The flagship McBurnie metric. Counts deceleration efforts per sprint effort. A player who decelerates many times per sprint is doing repeated braking — the most damaging single pattern. The threshold of concern is around 2.5 (ie. more than two and a half braking events per sprint).

This is the dimension MicroPulse weights most heavily in the overall verdict.

> **Indoor vs outdoor — automatic**
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> "Sprint" needs different definitions indoors versus outdoors. Outdoors we use Catapult Velocity Band 6+ effort count (gold-standard GPS sprint detection). Indoors GPS does not work, so we use IMA Free Running Bands 7+8 (high-intensity stride detection). The page picks the right denominator per session per player automatically — coach action is not required.

2.5 Concentration — "Was the load spread out or stacked?"

Looks at how many of the player's last 28 days of decel volume came from a single peak day. A player who got 60% of their 28-day decels in one session has a concentration problem — that one session was a spike, even if the 28-day average looks normal.

Acts on this when: a single matchday or hard session dominated the window. Often pairs with elevated ACWR.

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3. Reading the page

3.1 Top of page — counts grid

Four big numbers across the top:

- **Red** — players currently flagged on at least one dimension at the RED level
- **Yellow** — players with one or more YELLOW flags but no REDs
- **Green** — players within personal range on all five dimensions
- **Insufficient data** — players who don't have enough Catapult sessions yet to compute reliably (typically the first 14 days after onboarding)

Use this as the morning triage: you should be able to spot at a glance whether the squad is settled (mostly green) or whether several players are converging on overload (red and yellow climbing).

3.2 Decel-source banner

Just below the counts there's a small banner showing which Catapult field is currently feeding the McBurnie engine. It will say one of:

- **"Decel B3+ Total Efforts (Gen 2)"** — the most accurate source. This is the current default for clubs on Catapult Vector S7+ pods.
- **"IMA Decel Band 3"** — falls back to IMU-based detection when GPS efforts are unavailable.
- **"Mixed (B3+ where available, IMA Band 3 otherwise)"** — common during indoor weeks when the team has a mix of indoor and outdoor sessions.

Coaches should not need to act on this — it is purely transparent so you know which signal you are reading. If you ever see "Insufficient" here, it means Catapult is not returning the field at all — talk to your performance support.

3.3 Player rows

Each player gets a single row showing the 5 dimensions side-by-side as colored chips. The chip color (green / yellow / red / grey) tells you the status; clicking the row expands it to show the actual numbers underneath every chip.

Inside the expanded view you'll see, per dimension:

- The current value (e.g. "D:Sprint = 3.1")
- The player's personal baseline (e.g. "28-day mean: 1.8")

- The threshold that triggered the flag (e.g. “Threshold: ≤ 2.5”)
- A short plain-language explanation of what to do

Players are sorted RED first, then YELLOW, then GREEN, then “insufficient” — so the people who need your attention are always at the top.

3.4 Neuromuscular block (optional)

If your club has VALD ForceFrame, Nordbord or ForceDecks data flowing in, an additional **Neuromuscular** block appears at the bottom of each player’s expanded row. It shows:

- **Nordbord** — last hamstring eccentric strength test, and how many days ago it was done
- **ForceFrame** — last groin/hip strength test, and how many days ago it was done
- **ForceDecks (CMJ)** — last counter-movement jump, and the resulting capacity score

These complement the Catapult-based decel ratios with the player’s actual current capacity. A player flagged RED on D:Sprint who also has a fresh below-baseline Nordbord is much higher risk than the same flag on someone with strong recent strength tests.

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4. Workflow — how to use this in a normal week

4.1 Daily — morning briefing

Open the Decel Intelligence page first thing every morning, before the daily briefing meeting. Sort by RED. For every red player:

1. Click the row to expand
2. Read which dimension(s) are flagged
3. Look at the plain-language explanation
4. Decide modification: usually one of “skip the decel-heavy drill”, “drop to half-volume”, “swap into recovery group”

Most days you will have 0–2 red players on a healthy squad. If you see 4+ reds, look at *what’s common* — it’s usually the previous day’s session that was harder than planned, and the team-wide spike will show up here.

4.2 After every match

Decel Intelligence updates the morning after the match (once Catapult sync runs). Reading it post-match gives you the most accurate read on who absorbed the highest braking load — invaluable for the MD+1 / MD+2 modification call.

4.3 During return-to-play

For an injured player coming back, the dimensions to watch in order are:

1. **Underload first** — they will be far below match demand for weeks. Expected and OK.
2. **Concentration second** — once they're hitting volume, watch for a single spike day.
3. **A:D Coupling third** — when they start sprinting again, make sure decels stay matched.
4. **D:Sprint last** — once they're sprinting and decelerating freely, this becomes the readiness gate before full return.

The page does not replace clinical judgment from the medical team. It tells you what the player's body is *doing*; the medical team tells you what their tissue *can* tolerate.

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5. Buttons explained

There are three buttons in the top-right of the page:

- **↻ Re-sync síðustu 7 daga (Re-sync last 7 days)** — manually pulls the last 7 days of Catapult data again and recomputes. Use when you suspect Catapult was late uploading a session.
- **↻ Endur-reikna baselines (Recompute baselines)** — forces the personal baselines to refresh. Normally happens overnight automatically. Use after manual data corrections.
- **Greina Decel B3 (Diagnose Decel B3)** — runs a transparency probe against Catapult's API for today's data, showing which fields are returned and which are not. Used by the MicroPulse admin when investigating suspected data gaps. Coaches will rarely need this.

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6. Troubleshooting

"Insufficient data" — player has been on the squad for weeks

The personal baseline needs at least 14 days of valid Catapult sessions. If a player is showing "insufficient" after 14 days, check whether their Catapult pod is returning data at all (open /coach/load and search by player). Most common cause: the player is wearing a different pod number than the one assigned to them in OpenField.

Decel-source banner says "IMA Decel Band 3" all the time

This means Catapult is not returning the **B3+ Total Efforts (Gen 2)** field for your team. Either you are on older pods that don't compute it, or your reporting parameters in OpenField don't include it. Talk to your performance support — adding the parameter takes ~5 minutes in OpenField Cloud.

Same player keeps flagging red on D:Sprint week after week

Two patterns are common:

1. **Position effect** — central midfielders often have intrinsically higher D:Sprint ratios than wide players. Check their personal baseline (in the expanded row). If their personal baseline is ~3.0, a single week at 3.2 isn't actually a flag — the engine compares against personal range, but if you are reading the absolute number you might be over-reading. Trust the colored chip, not the raw number.
2. **Style effect** — some players genuinely play with more braking than the average professional. If they are healthy match after match, the elevated ratio is their normal. The Concentration dimension is the better risk indicator for these players.

A red flag I disagree with

The page is decision-support, not auto-pilot. If you have eyes-on context the data does not — player just had a personal issue, drill was abnormally short, etc. — your judgment overrides the model. The Calibration view on the main dashboard tracks how often coach overrides match what eventually happened, so over time you can see whether your model-overrides are systematically right or wrong.

Last updated: 28 April 2026 · MicroPulse Performance Intelligence Platform · Methodology: McBurnie 2022; Harper 2019; Jones 2024 (indoor IMA stride detection)